An Integrated Framework for Blood Rheology Analysis

UoM Group 35

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Client introduction

Haemograph

Innovative point of care blood coagulation rheometer empowering clinicians for enhanced anticoagulation therapy.

RheoCare™

Lightweight, portable, compact, point-of-care rheometer being designed to provide real-time accurate patient specific blood coagulation testing; incorporating easy loading of multiple reagents and providing easily understood diagnostics.

Learn More



Air Evacuation Syringe

Advanced syringe integrating our Autovalve™ technology to easily eliminate bubbles during loading without needing "tap-and-squirt", enabling the correct dose without waste.

Learn More

Autovalve™

A smart fluid check valve that vents gas while blocking liquid, automatically sealing upon fluid contact. Ideal for applications where bubbles must be removed during loading or priming, followed by precise control of liquid flow. Ideal for many industries.

Learn More

Revolutionising coagulation management.

Haemograph stands at the forefront of medical technology, dedicated to enhancing coagulation management. Our patented syringe, autovalve, and rheometer are designed to empower clinicians and researchers in Melbourne and beyond. Join us in advancing healthcare through cutting-edge solutions that make a difference.

More About Us

Group 35 members



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Yun Gu Literature Review Lead



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Project background and objective

Haemograph x THE UNIVERSITY OF MELBOURNE

Project background

- Collaboration with Haemograph
- Analyse how blood's mechanical behaviour changes over time
- Use rheology measurements to examine the response to applied deformation

Our objective

Identify and characterise changes in blood rheology measurements.

Explore what different analytical methods can tell us about these changes.

From clotting detection to change detection

The data do not include confirmed clotting times.
Observed changes therefore cannot be directly linked to clotting.

Data Overview

APR18

Experimental dataset used in this analysis

What we analyse

Stress response to applied shear deformation

LAOS

Large-Amplitude Oscillatory Shear

Repeated back-and-forth deformation at a large amplitude.

The resulting stress response is recorded.

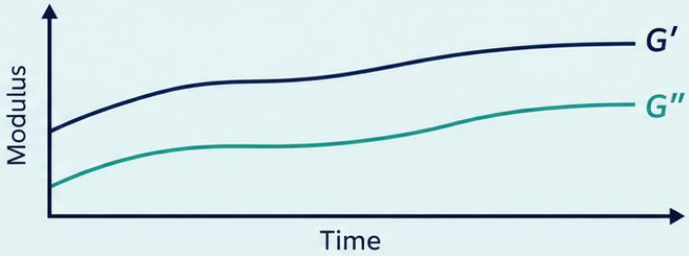
Synthetic data

Used for preliminary analysis of the methods.

Methods

Conventional analysis

G′: storage modulus — elastic response
G″: loss modulus — viscous response
tan δ = G″ / G′



Illustrative schematic

Physical modelling

t-ESSTV
Tensorial thixotropic viscoelastic model

Results and limitations discussed in the report.

Signal analysis

FFT	**Fast Fourier Transform** Shows the frequencies present in the signal.
STFT	**Short-Time Fourier Transform** Tracks frequency content using moving windows.
CWT	**Continuous Wavelet Transform** Examines local patterns across time and scales.
DWT	**Discrete Wavelet Transform** Separates slower trends and faster fluctuations.

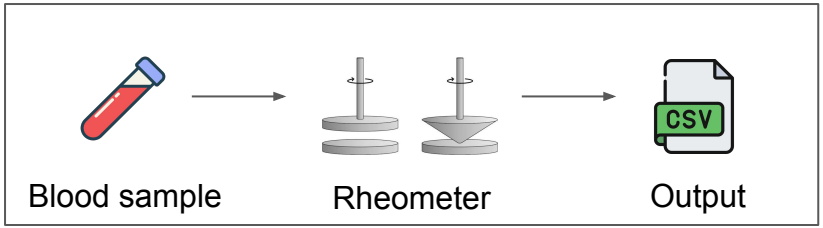
This presentation focuses on conventional and signal analysis.

What is Conventional Blood Rheology?

Blood is a **viscoelastic.**
Can be both liquid or elastic solid.

G' = G" gel-point (crossover point)
G' > G" elastic solid
G' < G" viscous liquid

How are measurements taken?



For Linear Small Amplitude Oscillatory Shear (SAOS)

Winter-Chambon Criterion

Loss tangent: $tan(\delta) = G'/G"$

For Non-Linear Large Amplitude Oscillatory Shear (LAOS)

Winter-Chambon Criterion analyzed with FFT cycles

Loss tangent: $tan(\delta) = G1'/G1"$

Conventional Analysis Results

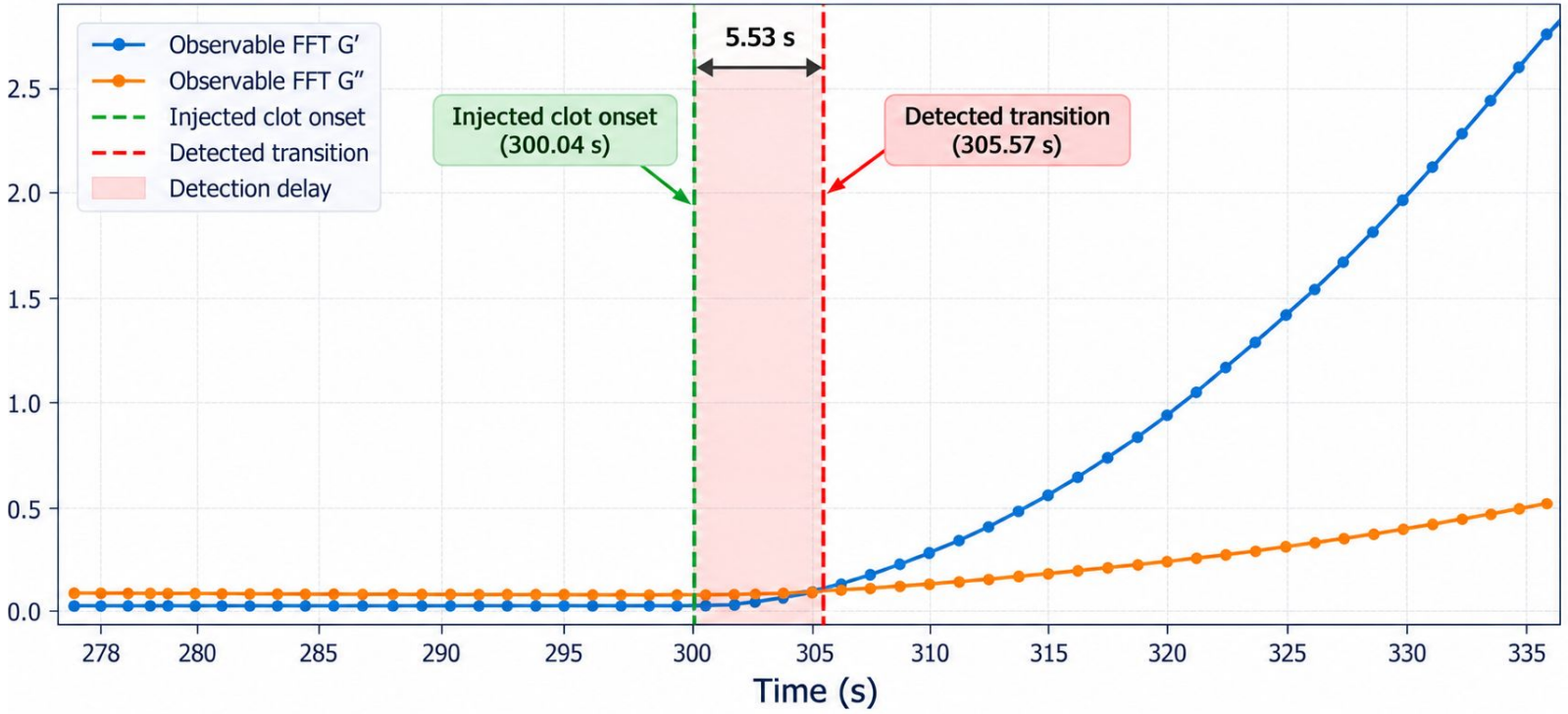


Figure 1: Synthetic clot_A crossover G' = G"

Conventional Analysis Results

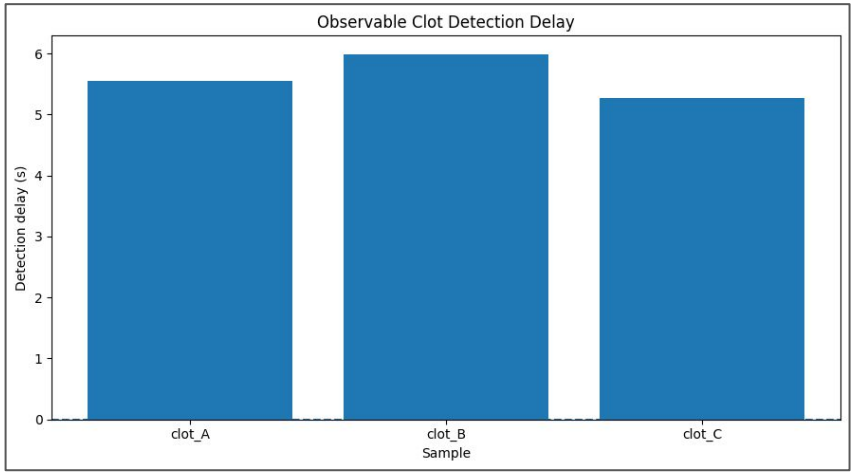


Figure 2: Clotting detection onset graph

Observed attributes:

- Consistent detection around 5-6s when clot is present
- Detection is delayed
- Assumes clot based on mechanical change

sample	datatype	status	Detection	Injected	Onset	FFT_Cycles
clot_A	LAOS	detected	305.57	300.04	5.53	1798
clot_B	LAOS	detected	246.01	240.04	5.97	1798
clot_C	LAOS	detected	365.29	360.04	5.25	1798

Figure 3: Clotting detection onset table

Why widely adopted conventional method?

Interpretable

Mechanical properties with clear physical meaning.

Established

A familiar framework for oscillatory rheometry.

Comparable

Summaries can be compared across times or samples.

Core measurements

G′

Storage modulus

Elastic energy stored

G″

Loss modulus

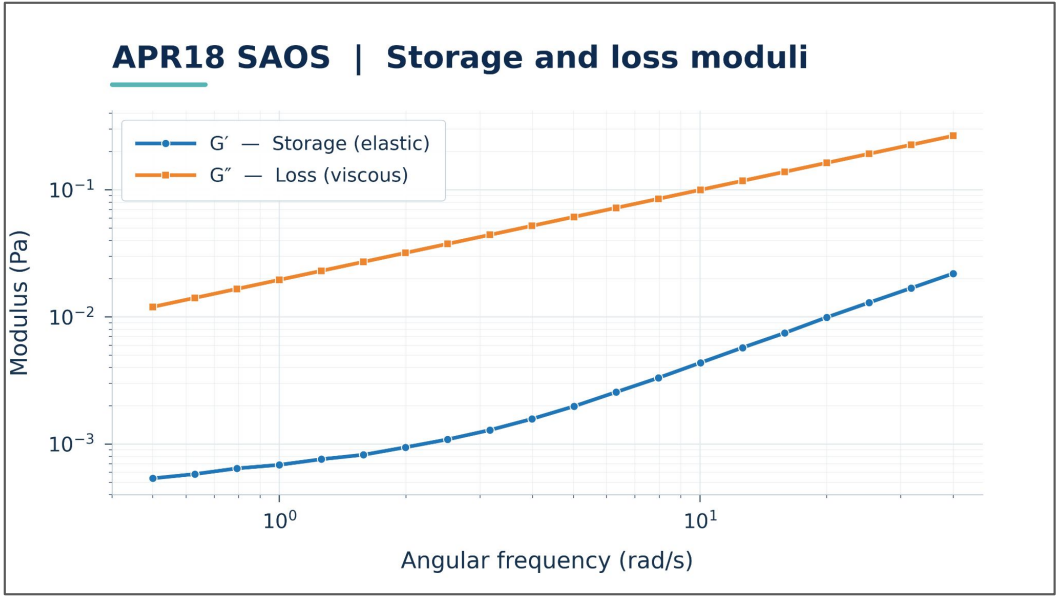
Viscous energy dissipated

tan δ

Loss / storage ratio

Relative viscous vs elastic

How conventional analysis may fall short



Important limitations

Timing is indirect

Crossover can lag an underlying change.

Cause is ambiguous

Mechanical changes alone do not identify the physical driver.

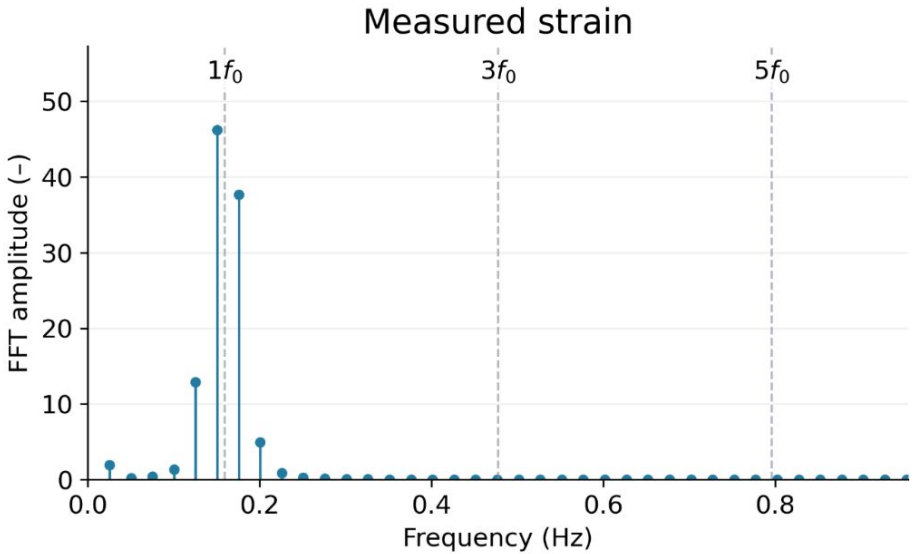
No confirmed clot times

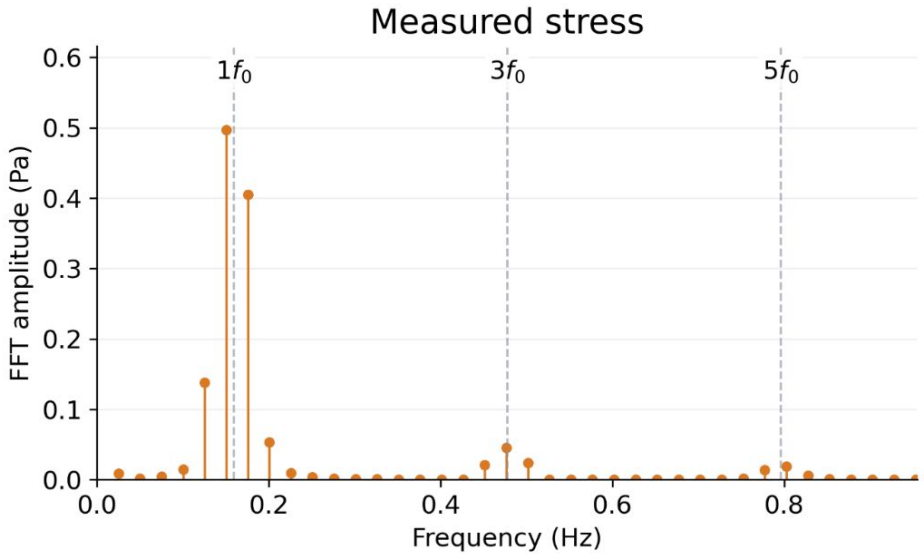
APR18 measurements do not provide ground-truth clot onset.

No crossover => No informative signals

Fast Fourier Transform reveals the harmonic

APR18 · LAOS · $\omega = 1$ rad/s · $\Gamma = 50$



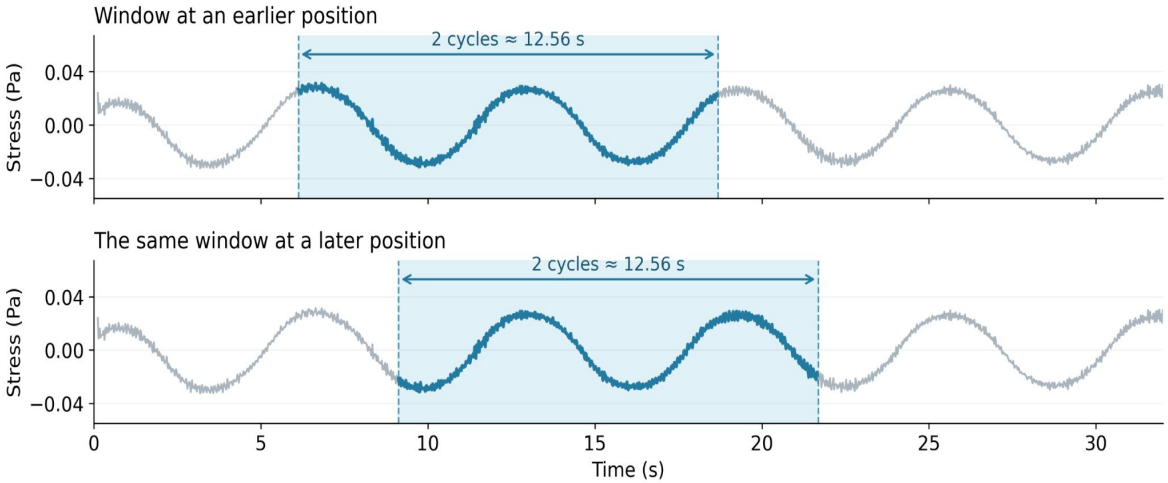


Amplitudes describe absolute strength. Ratios describe relative harmonic contributions.

The whole-record spectrum shows frequency content, but cannot locate changes in time.

Short-Time Fourier Transform windows and feature extraction

STFT: analyse a window, then move it along the signal



Window size: 2 oscillation cycles ≈ 12.56 s
The window moves along the measured stress signal.

Features	Information
I_1 , I_3 , I_5	Absolute harmonic amplitudes
I_3 /I_1 , I_5 /I_1	Relative harmonic contributions
Spectral power	Overall strength in the analysed band
Entropy / bandwidth	Power distribution / frequency spread

Each window produces eight features, which we track over time.

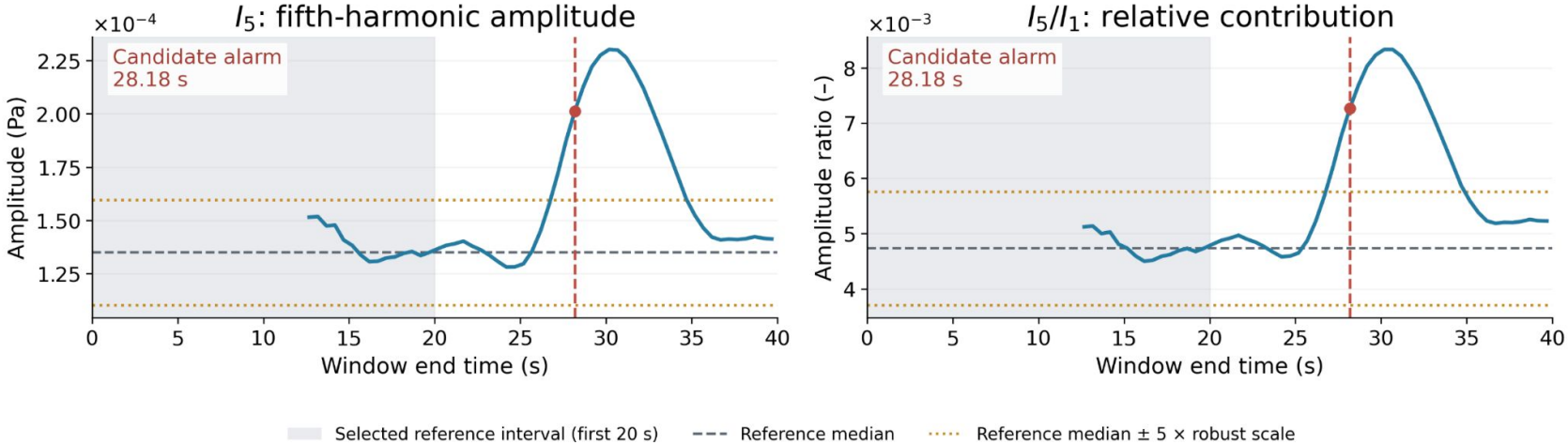
Departures from an initial reference

1 Select the first 20 s as an initial reference

2 Compare each feature with its reference level and variability

3 Confirm a same-side threshold exceedance lasting at least 1 s

APR18 · LAOS · $\omega = 1$ rad/s · $\Gamma = 0.5$

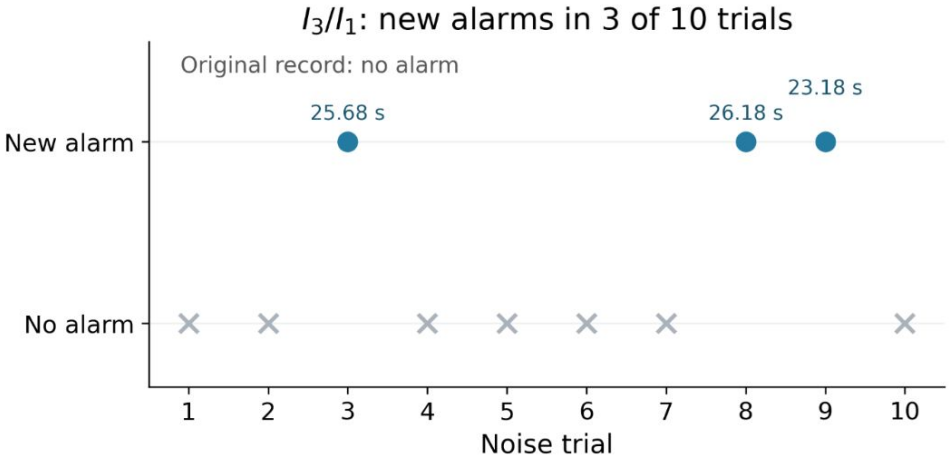


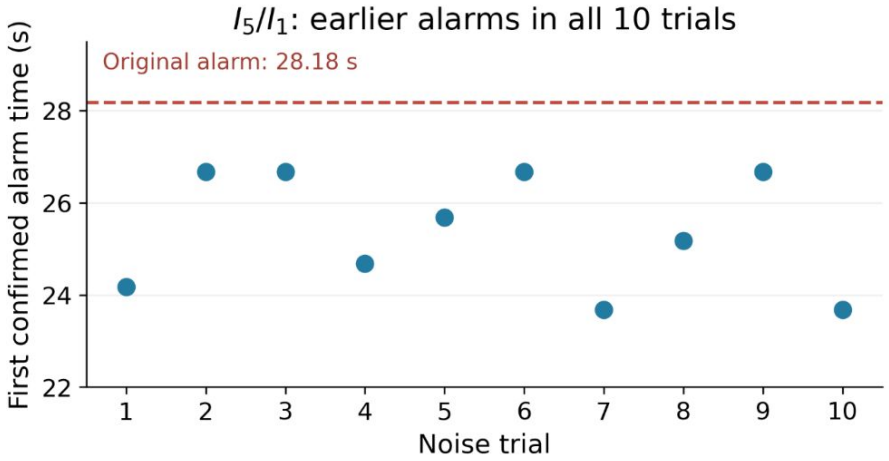
Both features flag a departure at 28.18 s. The physical cause remains unverified.

Noise changes STFT alarm outcomes

APR18 with added noise · $\omega = 1$ rad/s · $\Gamma = 0.5$

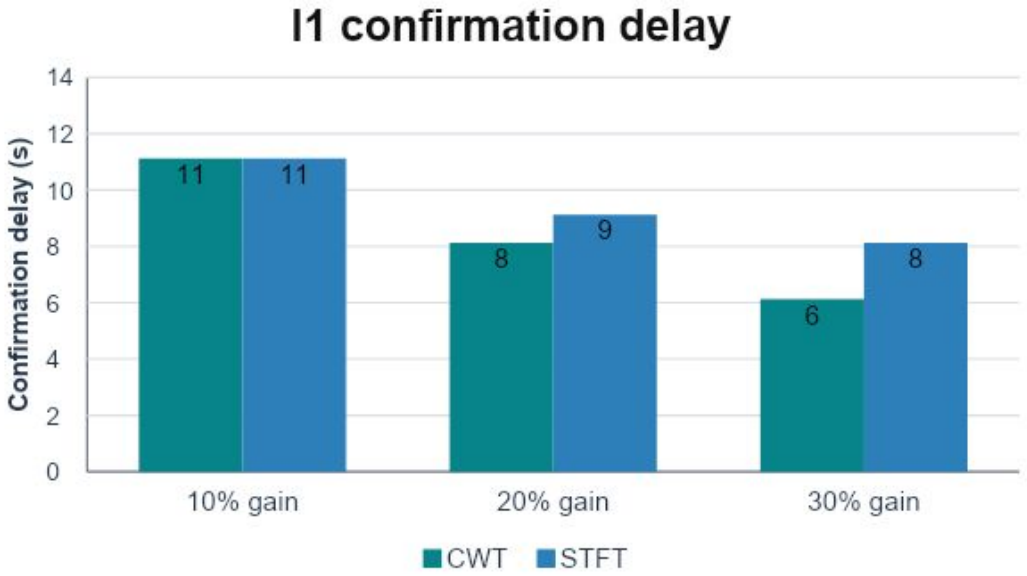
Noise added at 20–25 s · Noise RMS = 10% of pre-injection centred stress RMS





Added noise produces new alarms and shifts existing alarm times.

CWT: Detects faster in I1



Detection performance

I1 and Power confirmed all three gain cases.

CWT matched STFT at 10% gain.

CWT confirmed the 20% and 30% cases 1–2 s earlier.

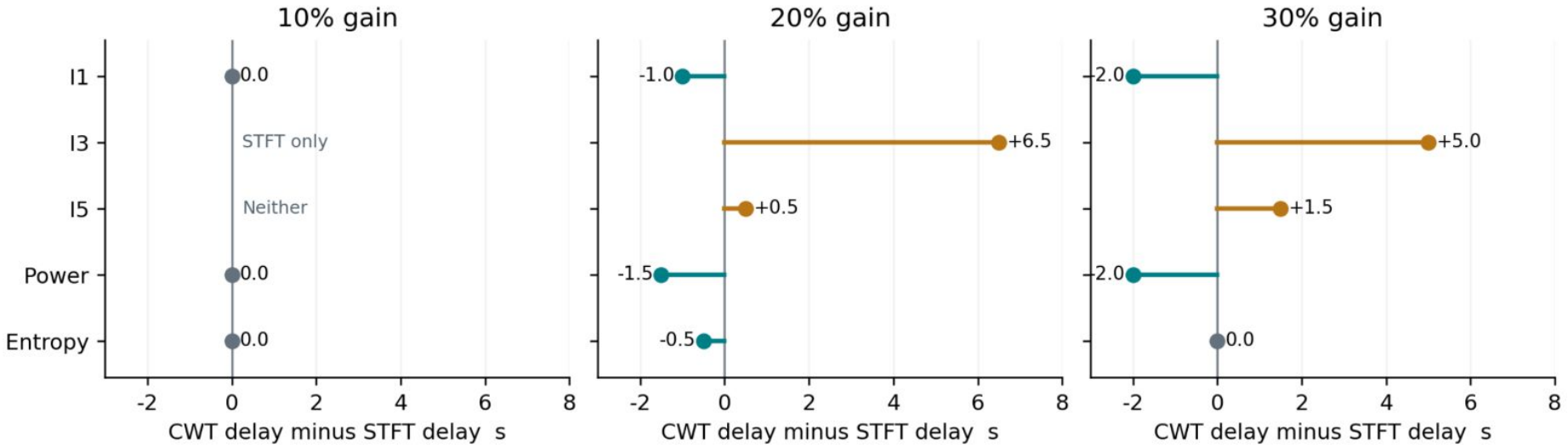
Feature-specific trade-off

STFT remained more responsive for I3 and I5.

CWT missed the 10% I3 change.

CWT performed best for fundamental amplitude and power, but its advantage depended on the feature.

CWT: Complete comparison with STFT



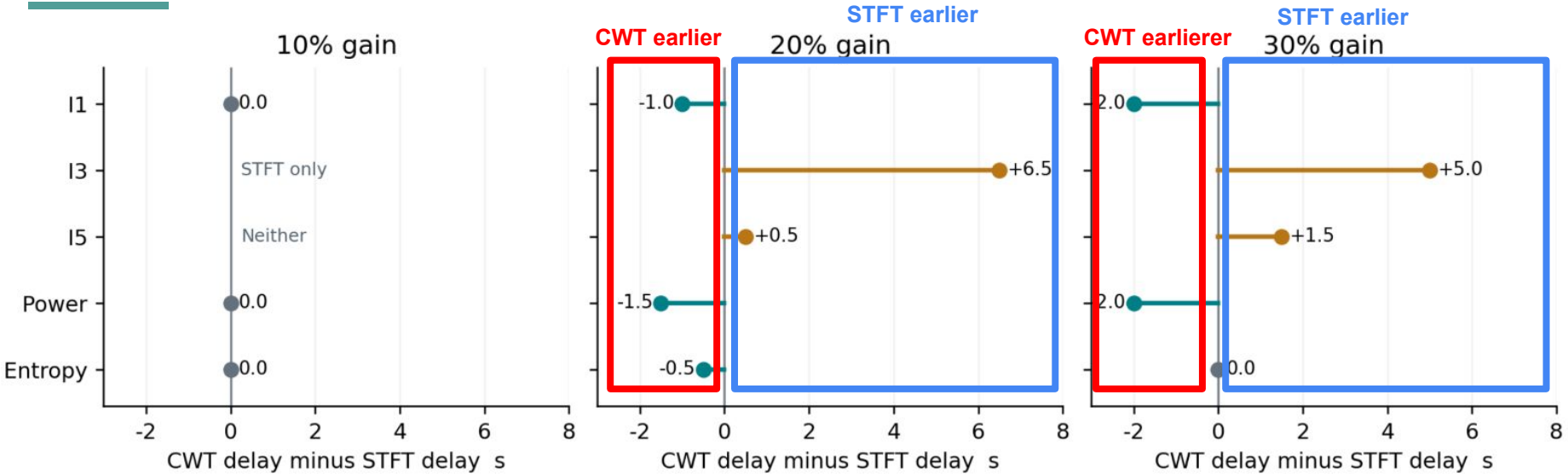
CWT: earlier confirmation for I1 and Power

STFT: earlier confirmation for I3 and I5

Neither: harmonic ratios did not respond to a common amplitude gain

CWT performed best for I1 and Power, while STFT was more responsive to higher harmonics.

CWT: Complete comparison with STFT

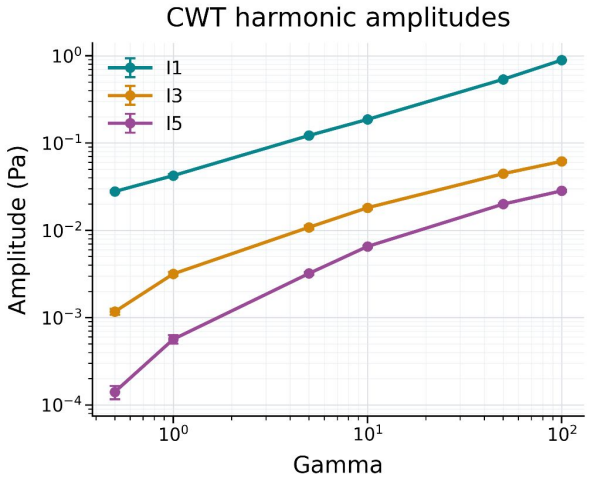


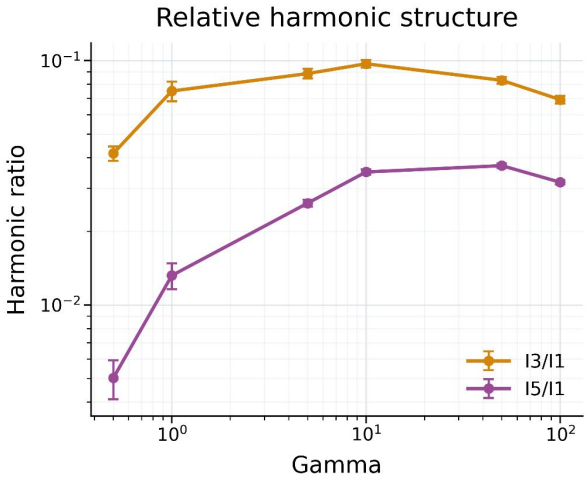
CWT: earlier confirmation for I1 and Power STFT: earlier confirmation for I3 and I5

Neither: harmonic ratios did not respond to a common amplitude gain

CWT performed best for I1 and Power, while STFT was more responsive to higher harmonics.

CWT results in the original APR18 data





Consistent harmonics

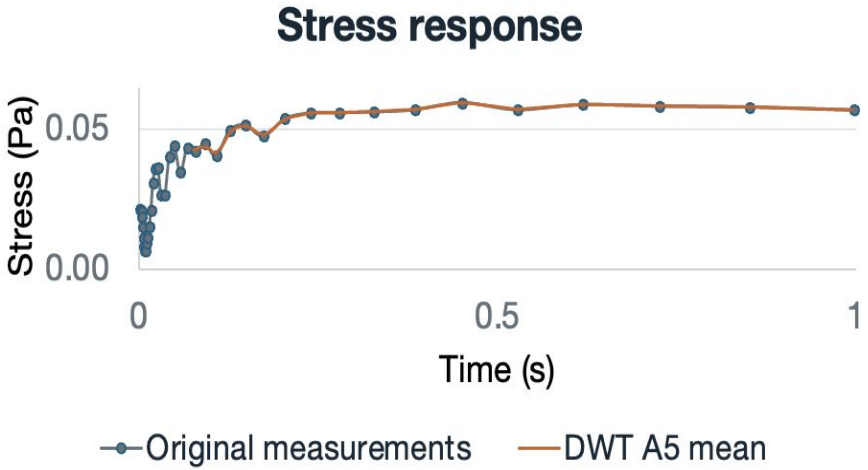
The fundamental, third and weaker fifth harmonics were present across all six records.

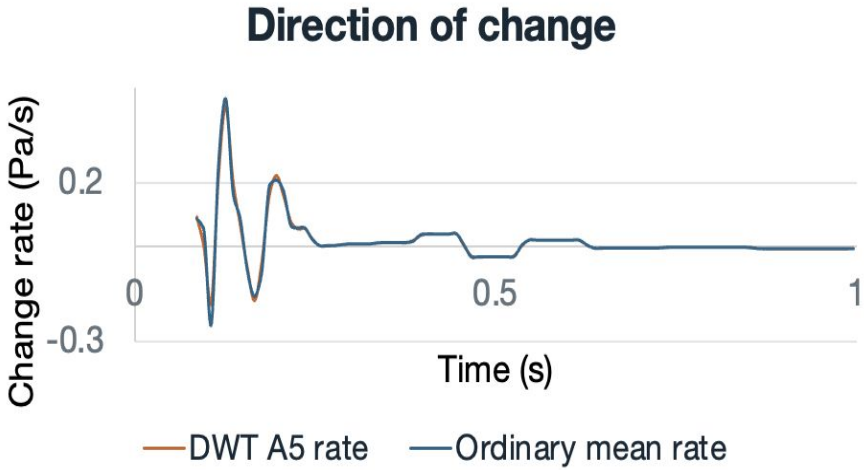
Amplitude dependence

Absolute I1, I3 and I5 amplitudes increased with Gamma, but their relative ratios were not monotonic.

Comparison	CWT	STFT
Fundamental and power	Matched at 10%; 1–2 s earlier at 20–30%	Same or later
Higher harmonics	Slower; missed 10% I3	Earlier for I3 and I5
Time–frequency analysis	Frequency-dependent scales	One fixed window
Known harmonics	More flexible, but more parameters	Direct and easier to interpret

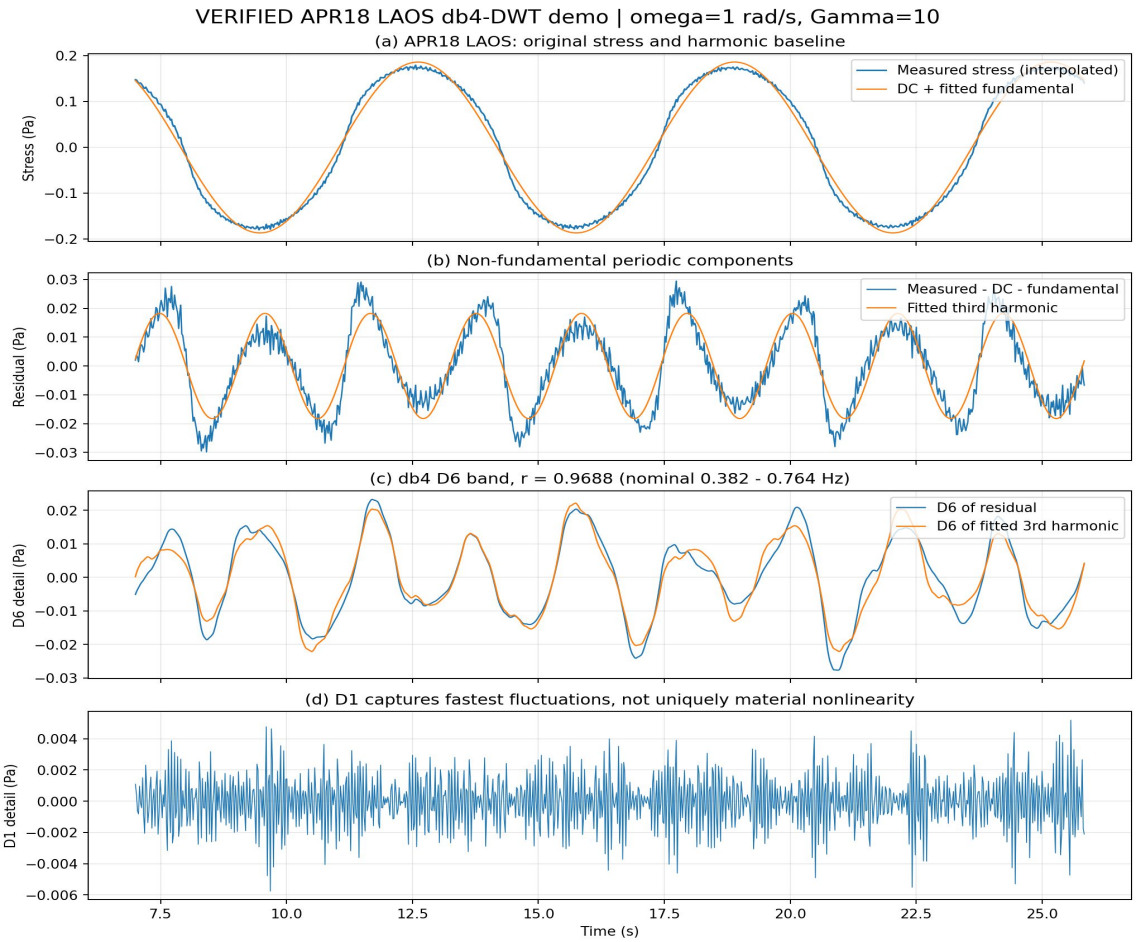
DWT: “STEP ”trend features describe the main response





Feature	Information extracted
A5 window mean (Pa)	Stress trend after DWT decomposition
A5 change rate (Pa/s)	Sign: rise / fall. Magnitude: speed of change.
Ordinary mean rate (Pa/s)	Non-DWT baseline: does DWT add useful information?

DWT: “LAOS” stress decomposition across scales



DWT can lift different frequencies from the signal using different scales.

(a) Main stress response
Stress follows the loading cycle, with deviations from a sine wave.

(b) Remaining pattern
After removing the basic rhythm, a repeating third harmonic is visible.

(c) D6 comparison
Residual D6 and fitted-third D6 have similar shapes: $r \approx 0.969$.

(d) Fast D1 fluctuations
Fast activity alone does not confirm noise or a change point.

Conclusion

Criterion	Conventional	FFT	STFT	CWT	DWT
Analyses	G′, G″, tan δ	Frequency spectrum	Moving-window spectra	Time-frequency patterns	Trends and details
Main finding	Viscoelastic behaviour	Odd harmonics	Candidate departures	Persistent harmonic bands	STEP trend; third harmonic
Time localisation	By analysis window	No	Yes	Yes	Yes, scale-dependent
Main limitation	Cause unclear	No timing information	Noise-sensitive	Feature-dependent	Processing-sensitive; physical meaning unclear

Data Science Contributions

01

Reusable workflow

A structured workflow for analysing blood rheology measurements.

02

Feature extraction

Rheological, frequency and time-scale features from complementary methods.

03

Method evaluation

Synthetic data and added-noise experiments to examine method responses.

04

Practical outputs

Code and documentation to support Haemograph's future analysis.

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Thank you！😊 Q&A